

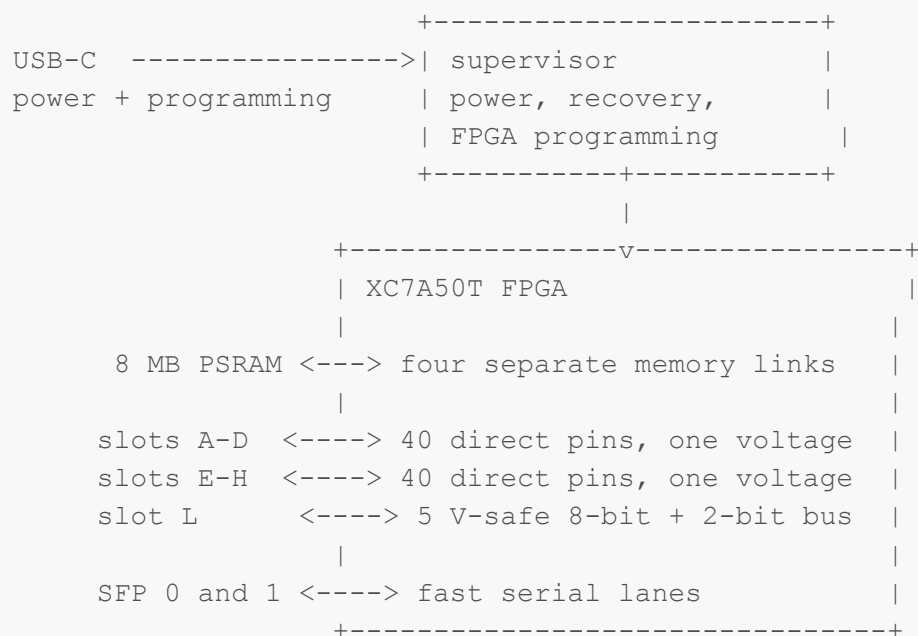
Odin user manual

Odin is an FPGA board for building the thing you meant to build.

One USB-C cable powers it and programs it. Four PSRAM chips give the FPGA 8 MB of working memory. Nine extension slots carry signals and useful power. Two SFP sockets carry the links that ordinary FPGA pins cannot.

This manual is for using the board. If you are designing an extension, read [ref/extension-ux.md](#) after this.

The board in one picture



Start here

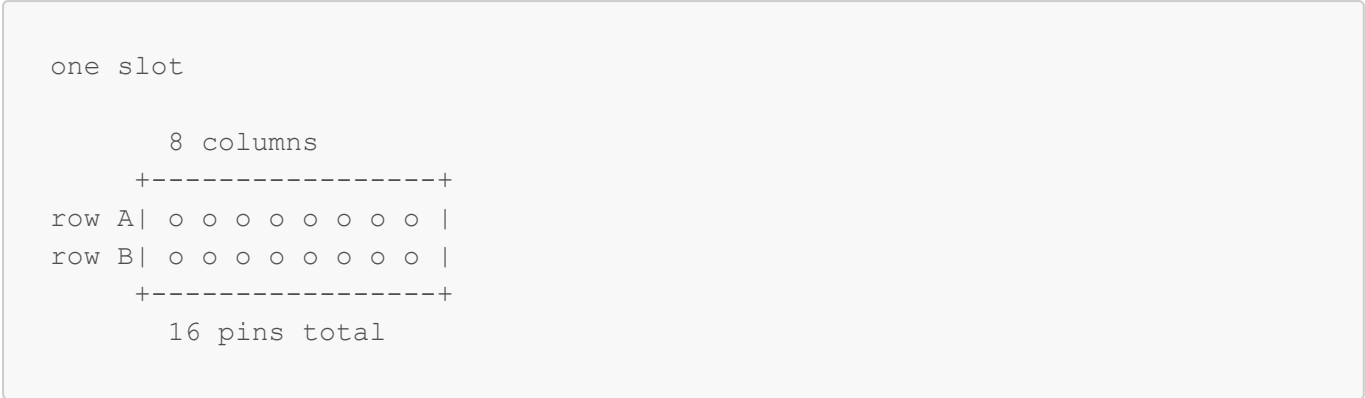
1. Leave all extensions unplugged.
2. Set both VCCIO selectors to **3.3 V**.
3. Set each AUX selector to **3.3 V**, or leave its jumper off.
4. Plug USB-C into Odin.
5. Wait for the supervisor to start.
6. Plug in one simple extension and test it before adding the next.

Use a USB-PD charger when the board carries displays, speakers, motors or SFP modules. A computer port is fine for programming and light work, but it may not have enough power for everything at once.

Odin never asks USB-C for more than 9 V.

The extension idea, from zero

Odin has eight shield connectors arranged in two matching rows on opposite sides of the 100 x 100 mm board. Each connector has 16 metal pins. A ninth connector, L, is the separate 5 V-safe legacy slot.



Ten of those pins are signals. The other six are ground and power.

10 signal pins + 2 ground pins + 4 power pins = 16 pins

Where are the I/Os?

Here. IO means a signal you can use as an input or an output.

	1	2	3	4	5	6	7	8
row A	IO0	IO2	GND	IO4	IO6	IO8	VCCIO	+5V
row B	IO1	IO3	GND	IO5	IO7	IO9	+3V3	AUX
	<--- signals --->			<--- signals --->			power	power

The ten I/Os are columns 1, 2, 4, 5 and 6. Each of those columns has two signal pins. Column 3 is ground. Columns 7 and 8 are power.

The schematic calls the same signals P0+, P0-, and so on. You do not have to use them as pairs:

connector name	ordinary use
P0+ and P0-	IO0 and IO1
P1+ and P1-	IO2 and IO3
P2+ and P2-	IO4 and IO5
P3+ and P3-	IO6 and IO7
P4+ and P4-	IO8 and IO9

They are called pairs because the FPGA can send one fast signal over two wires. One wire rises while the other falls. This rejects noise. If you do not need that, use them as ten ordinary, unrelated I/Os.

What do 16, 32 and 64 mean?

They mean the width used on **each side** of the base board. An extension is a bridge, not a board hanging from one edge.

	one edge of Odin		opposite edge
16 shield	[A]	=====	[E]
	16 pins	extension	16 pins
32 shield	[A][B]	=====	[E][F]
	32 pins	extension	32 pins
64 shield	[A][B][C][D]	=====	[E][F][G][H]
	64 pins	extension	64 pins

A 16 therefore connects to one 16-pin header on each side: 32 physical pins and 20 I/Os in total. A 32 connects 32 pins to 32 pins. A 64 connects all 64 pins on one side to all 64 on the other: 128 physical pins and 80 I/Os.

The schematic names four 16-pin sections on each edge. Those are electrical zones, not a demand for four separate plastic connectors. Odin can use the original physical arrangement: **two continuous 32-pin stack-through headers on each edge**. A 16 shield uses half of one header, a 32 uses one whole header, and a 64 uses both headers. The keying and outlines must make those partial positions unambiguous.

The bridge gives the extension support at both ends, almost the full board area above Odin, and matching connectors on top so another shield can stack. The extension PCB and its fitted headers determine its size; there is no size switch.

Where are the slots on Odin?

The required arrangement is:

```

                        100 mm
+-----+
| [ A ][ B ][ C ][ D ]           | bank 15
|                               |
|           ODIN                | shield bridges
|                               | across this space
| [ E ][ F ][ G ][ H ]           | bank 34
+-----+

[ L ] separate 5 V-safe legacy connector

```

Matching pairs are A/E, B/F, C/G and D/H. Placement must put each pair directly opposite its partner. All eight connectors need the same pitch, orientation and distance from the board edges.

Their exact coordinates do not exist yet because placement has not been done. Until those coordinates and a real extension outline are drawn and measured together, the bridge and stacking fit remain an intention, not a proven feature.

Slots A-D always share one signal voltage. Slots E-H share another. Changing a VCCIO jumper changes the voltage of the whole group, not one slot.

Shield	Connectors underneath	Pins per side	Total pins	Usable I/Os
16	A+E, B+F, C+G or D+H	16	32	20
32	Two matching pairs	32	64	40
64	A-D and E-H	64	128	80

A shield reaches both FPGA banks. The A-D side shares one VCCIO setting; the E-H side shares the other. A 64 owns both complete groups. A smaller shield must share each side's voltage with anything using the unused connector pairs.

Stacking

Every shield needs female sockets underneath and matching male pins on top. The signals pass vertically through the board so another shield can plug in.



Stacking does not create more FPGA pins or more power. Stacked shields share the same wires, voltage groups and current budget. They must not drive the same signal at the same time.

The key posts, stacking headers, component height and board-to-board spacing must be designed as one mechanical system. A key that works for one board but blocks the next board is not a working key.

One slot, using the schematic names

	column	1	2	3	4	5	6	7	8
row A		P0+	P1+	GND	P2+	P3+	P4+	VCCIO	+5V
row B		P0-	P1-	GND	P2-	P3-	P4-	+3V3	AUX
		signal pairs			signal pairs			power pins	

The separate post beside the header is the key. If the extension does not drop onto the header and key without force, it is the wrong way round.

Never shift the extension sideways by one column. Check the key before power.

What the four power pins mean

Pin	Meaning
VCCIO	The signal voltage for that FPGA bank: 3.3, 2.5 or 1.8 V
+5V	Fixed supply for 5 V parts
+3V3	Fixed supply for ordinary 3.3 V parts
AUX	Chosen beside the slot: 12 V, 5 V or 3.3 V

VCCIO is both a power output and the voltage used by the FPGA pins. That is why the signals and the extension cannot silently disagree.

The 1.0 V FPGA core rail never reaches a connector.

Power is shared

The small fuse beside each slot is fault protection. It is not a promise that every slot may draw the number printed on the fuse at the same time.

The 12 V AUX rail is the tightest. Treat **about 0.5 A across the whole board** as the present design target, pending final component qualification. One LCD backlight is sensible. Nine motors are not.

The final production manual must take its exact limits from the released power-budget report. Do not copy preliminary figures from an old schematic.

Which slot should I use?

```
Is the signal 5 V?
```

```
yes -> use L
```

```
no  -> continue
```

```
Does each pin need its own direction, or is it fast/LVDS?
```

```
yes -> use A-H
```

```
no  -> either works
```

```
Does it require 1.8 V or 2.5 V?
```

```
yes -> give it a whole bank: A-D or E-H
```

```
no  -> leave VCCIO at 3.3 V
```

Slot L is different

Slot L protects the FPGA from 5 V signals. It is not ten unrelated GPIOs.

```
I/O 0-7 : one 8-bit bus, one shared direction
```

```
I/O 8-9 : one 2-bit bus, one shared direction
```

All eight pins in the first group turn around together. The last two turn around together. Use L for old 5 V buses, parallel displays and similar parts. Do not use it for I2C, one-wire parts, or a mixture of inputs and outputs that must change direction independently.

Slots A-H connect directly to FPGA pins. They are flexible and fast, but they are **not 5 V tolerant**.

Example 1: a 3.3 V SPI sensor

Use one direct slot, for example A.

```
Sensor VCC    -> +3V3
Sensor GND    -> GND
Sensor SCK    -> any signal pin
Sensor MOSI   -> any signal pin
Sensor MISO   -> any signal pin
Sensor CS     -> any signal pin

VCCIO group A-D -> 3.3 V
AUX jumper      -> not needed
```

Six signal pins remain free. The sensor receives clean 3.3 V even if another bank uses a different voltage.

Example 2: an old 5 V parallel peripheral

Use slot L.

```
Peripheral VCC      -> +5V
Peripheral GND      -> GND
Data D0-D7          -> slot L I/O 0-7
Read/write or status -> slot L I/O 8-9
```

In the FPGA design, drive the 8-bit direction control before driving or reading D0-D7. Do the same separately for I/O 8-9.

Do not connect a 5 V signal to A-H just because its power comes from their 5 V pin. Power voltage and signal voltage are separate things.

Example 3: a small handheld

A 64 shield bridges A-D to E-H and gets eighty direct 3.3 V signals. This example only needs thirty-seven:

```
16 LCD colour bits
4  pixel clock and display timing
8  buttons
```

```

2  display reset and backlight control
3  digital audio
4  microSD SPI
---
37 used
43 spare

```

Use:

```

VCCIO  3.3 V   FPGA signal level
+3V3           display, SD card and audio logic
+5V           audio amplifier
AUX      12 V   backlight driver, if required

```

The framebuffer belongs in PSRAM. A 320 x 240 RGB565 frame is about 150 KB; the board has 8 MB.

Memory without ceremony

Odin has four 2 MB PSRAM chips. Each has its own connection to the FPGA.

```

PSRAM 0  video or DMA
PSRAM 1  CPU memory
PSRAM 2  CPU memory
PSRAM 3  scratch, capture or another CPU

```

That split is only an example. HDL decides what each chip does. The hardware does not force them to share one busy connection.

One chip can move roughly 50 MB/s at a 100 MHz quad-SPI clock. Four used together can approach 200 MB/s. VGA RGB565 fits on one. 1080p60 RGB565 does not fit on all four. That is the honest ceiling.

Programming and recovery

The supervisor has its own flash. The FPGA has another. They are not the same thing.

Storage	Holds
Supervisor flash	The software that runs the board
FPGA flash	FPGA images and fabric software

A new blank board enters its built-in USB loader. Copy the supervisor firmware to the USB drive that appears.

If a later supervisor update goes wrong:

1. Unplug power.
2. Hold **BOOTSEL**.

3. Plug USB-C back in.
4. Release the button when the USB drive appears.
5. Copy known-good supervisor firmware to it.

The recovery loader is inside the chip. A broken flash image cannot erase it.

External JTAG

Normal use needs no external programmer. J2 exists for low-level FPGA debug. Fit the JTAG isolation jumper before attaching an external programmer; that disconnects the supervisor from the same four wires. Remove the jumper for normal supervisor-controlled programming.

Fast links

Two SFP sockets are fitted by default. They can accept suitable copper, fibre or direct-attach modules.

```
SFP 0 -> FPGA fast lane 0
SFP 1 -> FPGA fast lane 1
```

The FPGA provides the physical serial lane. Your FPGA design still needs the protocol logic. Plugging in a copper Ethernet module does not create an Ethernet controller by itself.

The lanes top out around 6.6 Gb/s. Use them for 1 Gb/s or suitable 2.5 Gb/s links, not 10 Gb/s modules.

Two more unfilled SFP positions preserve lanes 2 and 3 for a later build.

If something goes wrong

The board resets when an extension starts

The power source is probably too small, or the extension has too much startup current.

- Remove the extension and try again.
- Use a USB-PD charger instead of a computer port.
- Check the extension for a short.
- Start with AUX disabled or set to 3.3 V.

An extension does nothing

- Check the key and header alignment.
- Check the VCCIO setting for the whole bank.
- Check its AUX selector.
- Remember that Slot L uses shared direction controls.
- Remember that a blown or tripped branch fuse is not fixed by HDL.

The FPGA will not program

- Remove every extension.
- Restore both VCCIO groups to 3.3 V.

- Remove the external-JTAG isolation jumper for normal programming.
- Recover the supervisor with BOOTSEL.

A rail reads low

Disconnect extensions one at a time. A bad extension should be treated as the fault until it works alone on a current-limited bench supply.

Rules worth remembering

1. Check the key before power.
2. Slots in one bank share one VCCIO setting.
3. Only Slot L accepts 5 V signals.
4. Slot L is an 8-bit bus plus a 2-bit bus, not ten independent GPIOs.
5. The four connector rails share the board's total power.
6. Never feed power into an Odin power-output pin.
7. Start a new extension at 3.3 V with AUX disabled.
8. Test one extension before stacking several.
9. Use a PD charger for power-heavy projects.
10. When in doubt, unplug power first.

Before this manual becomes a release manual

The circuit is still being qualified. Replace this section with final ratings before shipping hardware:

- Exact shared and per-slot current limits
- Extension-rail shutdown and fault behavior
- Final connector positions and orientation drawing
- Supervisor status messages and firmware-loading names
- Qualified SFP module types
- Board revision and matching schematic commit

Until those are fixed, this is the user contract for the design—not permission to fabricate an unchecked revision.